

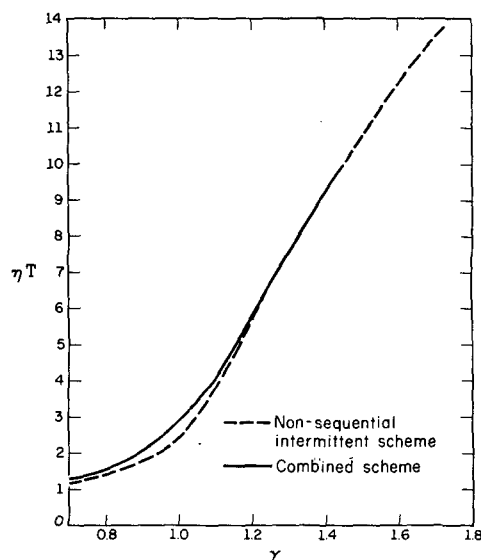


Fig. 4. Normalized peak power versus normalized average energy per symbol ($N = 10$).

be noted that the difference in peak powers between the combined scheme and the nonsequential intermittent scheme is very small.

V. SUMMARY AND CONCLUSIONS

In this paper an intermittent feedback communication scheme for binary constant antipodal signals is proposed. This is a variable length coding scheme where the final decision is made at a random time not exceeding T , depending on the uncertainty of the receiver. The forward channel is assumed to be disturbed by additive white Gaussian noise and the feedback channel is noiseless.

Expressions for the average decision time, probability of error, average power, and the normalized average energy per symbol have been derived. It has been shown that the proposed scheme, which is a combination of the intermittent non-sequential and the sequential schemes, achieves a better performance especially in the region where the signal-to-noise ratio is small. The combined scheme becomes Kramer's [7] intermittent feedback scheme or the sequential scheme with information feedback as Y or T approach infinity, respectively. It is seen that the improvement is achieved at the expense of increased signal peak power and extra system complexity due to an additional feedback channel.

However, the combined scheme discussed here cannot be used for transmitting M level signals ($M > 2$). Also, it is not clear how this scheme would perform if nonconstant signals are transmitted.

ACKNOWLEDGMENT

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Coherent Demodulation of Frequency-Shift Keying With Low Deviation Ratio

RUDI DE BUDA, MEMBER, IEEE

Abstract—A coherent binary FSK modulation system is discussed, that has the following properties: 1) it is phase coherent; 2) it has a low deviation ratio, $h = \frac{1}{2}$; 3) it occupies a small RF bandwidth, typically only 0.75 times the bit rate, without need for intersymbol interference correction; 4) it uses as receiver a self-synchronizing circuit and a phase detector, which together achieve optimal decisions; and 5) its error performance is about 3 dB better than that of conventional FSK.

I. INTRODUCTION

This paper describes a scheme to transmit binary signals over an RF link.

Usually such digital signals are encoded either into frequency-shift keying (FSK) or phase-shift keying (PSK). While FSK has the advantage of simpler circuitry, it is generally conceded that PSK has the advantage of better utilization of bandwidth and signal-to-noise ratio.

It will, however, be shown in this paper that the coherent FSK with the deviation ratio 0.5 has an optimal detector, which is also simple to construct. Using this detector, one can receive faster pulse trains than with any other binary FSK or PSK of equal bandwidth and signal-to-noise ratio.

II. PHASE COHERENCE

We consider a phase-coherent binary frequency-shift-keyed (FSK) signal and call a pulse at frequency f_1 a "mark" and at $f_2 > f_1$, a "space." Each pulse has duration T .

If the signal is narrow band, e.g., at IF or RF, then we may describe it by its pre-envelope [1] $u(t)$;

$$u(t) = \exp j\{\pi(f_1 + f_2)t + \phi(t)\}, \quad (1)$$

where the phase function $\phi(t)$ is continuous if the FSK is coherent.

Comparison of (1) with the pre-envelope of a space

$$u_{\text{space}} = \exp j\{2\pi f_2 t + \phi(0)\}, \quad 0 \leq t \leq T$$

gives

$$\phi(T) - \phi(0) = \pi(f_2 - f_1)T. \quad (2)$$

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The author is with the Canadian General Electric Company, Ltd., Toronto, Ont., and McMaster University, Hamilton, Ont., Canada.

For a mark, the phase decreases by this same amount. This suggests introducing, by definition, the dimensionless parameter h ;

$$h = (f_2 - f_1)T, \quad (3)$$

which is called modulation index [2] or frequency deviation ratio.

A space increases the phase by πh and a mark reduces it by πh . After s spaces and m marks, i.e., at time $(s + m)T$, we have then the phase shift [3, fig. 1]

$$\phi\{(s + m)T\} - \phi(0) = (s - m)\pi h, \quad (4)$$

which is an even (odd) multiple of πh when $(s + m)$ is even (odd). The phase at all other times is obtained by linear interpolation. The possible values of $\phi(t)$ are shown in Fig. 1.

Since the phase $\phi(t)$ is a continuous function of time, the signal itself is continuous and has thus lower spectral sidelobes than a pulsed signal. Consequently, a coherent FSK with low deviation ratio h occupies less bandwidth than other comparable systems. This has been noticed for some time [2]–[6] and will be utilized here.

III. FSK WITH LOW MODULATION INDEX h

From here on we will restrict our attention to coherent FSK with small modulation index h and in particular to the three cases

$$h = 1.0$$

$$h = 0.71$$

$$h = 0.5.$$

The FSK with $h = 1.0$ has been described in detail by Sunde [6]; it has not found wide acceptance. It uses signal power inefficiently because it transmits some power at the spectral lines f_1 and f_2 [7]. More on this topic will be found in Section VII.

The case $h = 0.71$ has been claimed to be optimum [8], [9] under certain conditions, which will be discussed later. This system has been extensively studied by Thjung, who shows in particular that, in a very narrow band coherent FSK with $h = 0.7$ is better than phase-shift keying PSK, because the FSK has lower intersymbol interference [10].

The system that will occupy most of this report is the coherent FSK with deviation ratio of exactly $h = \frac{1}{2}$. It has a spectrum that falls off smoothly but rapidly [3, fig. 2], [4, fig. 4] and it has been singled out as a good example of band conservation [5], [11]. Apart from this, the superior qualities of this modulation seem to have attracted little attention until very recently [12]–[14] and one finds often that PSK is accepted as superior to FSK [15], [16].

After some comments on the so-called optimum at $h = 0.71$, the coherent FSK with deviation ratio $h = 0.5$ will be taken up again. First its optimal decision structure will be given, which requires a coherent demodulation; and self-synchronizing circuits will then be described, which are essential to obtain the optimal performance. With such a receiver, the signal requires less bandwidth than binary PSK and less power than any other binary FSK and comes close to four-phase PSK, which has much more complex receiver synchronization requirements. These advantages might be indicated by dubbing this FSK modulation the "fast FSK."

IV. DISCUSSION OF THE OPTIMIZATION OF h

The statement that this fast FSK is better than all other FSK seems to be at variance with the accepted value of $h = 0.715$... for optimum error probability. Therefore, it will be instructive to repeat here the derivation of the optimum at

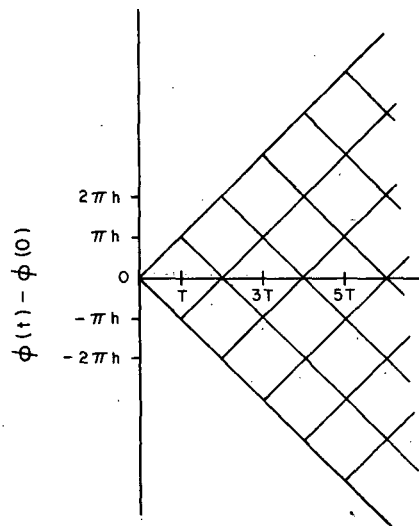


Fig. 1. Possible values $\phi(t) - \phi(0)$.

$h = 0.715$, so that we can see more readily where a false restriction has been applied [17].

We assume, that at time $t = 0$ the phase of the received signal is given, say $\phi(0)$, and that for $0 \leq t \leq T$ either space or mark is transmitted.

We assume further, that the IF center frequency is sufficiently high. Then the narrow-band assumption is valid, so that the signals can be expressed by their preenvolopes;

$$u_{\text{mark}} = \exp j[2\pi f_1 t + \phi(0)]$$

or

$$u_{\text{space}} = \exp j[2\pi f_2 t + \phi(0)].$$

Next, white Gaussian noise is added to the signal. Then a decision must be made at the receiver as to whether mark or space has been transmitted (see Fig. 2).

The lowest error of this decision is obtained when the distance between the two signals in Hilbert space is largest [18]. Since mark or space are transmitted only for $0 \leq t \leq T$, one evaluates thus first the integral

$$D^2(h) = \int_0^T |u_s - u_m|^2 dt \quad (5)$$

and then maximizes $D(h)$ with respect to h .

Simple substitution gives, with (3),

$$|u_s - u_m| = 2 \sin \frac{\pi h t}{T}$$

and

$$D^2(h) = 4 \int_0^T \sin^2 \frac{\pi h t}{T} dt = 2T[1 - \text{sinc } 2h]. \quad (6)$$

Woodward's sinc function [1, p. 29] is known to have its first and lowest minimum (-0.217) when

$$\tan 2\pi h = 2\pi h. \quad (7)$$

The maximum of $D(h)$ occurs therefore when $h = 0.715$ so that this is assumed to be the optimum value for h .

The fallacy in the preceding derivation lies in the arbitrary selection of the limits $0 \leq t \leq T$ in the integral that defines the distance $D(h)$, rather than integrating over $(-\infty, \infty)$. The same applies to the experimental work, which seems to confirm the theory; but bases the decision also only upon the

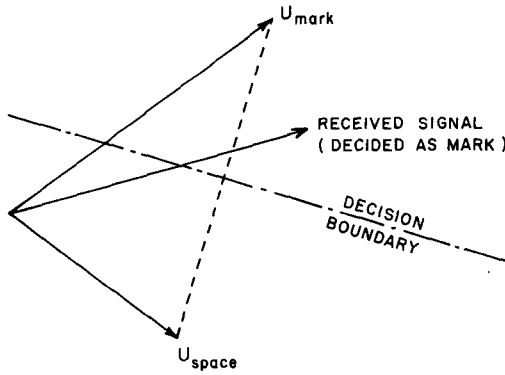


Fig. 2. Hilbert space representation of the mark-space decision.

interval $(0, T)$. This is inadequate, because the interval $(0, T)$ is not the complete time that is available to make the decision. It is true that T is the duration of the transmission of one bit and we must make one decision per transmitted bit; but the bit phases are interrelated and it is a basic tenet of information theory that better performance is obtained when several signals are jointly investigated over a longer period. Now low modulation index coherent FSK retains useful phase information beyond the time of the bit transmission [15]. In the binary case this extends over two time intervals. (For suitably selected m -ary FSK, the phase coherence can extend over several time intervals, but this will not be pursued here.) In particular, the phase in the succeeding time interval, $T \leq t \leq 2T$ contains unused information about the transmitted frequency of the earlier time interval $0 \leq t \leq T$; this information increases when h decreases (below 0.71) and becomes particularly important at $h = 0.5$. A receiver that fails to use this information will therefore achieve at best a local maximum, but not an optimum, and thus we should expect a receiver with lower h and a decision interval $2T$ to perform better than the so-called optimum at $h = 0.715$.

V. OPTIMUM RECEPTION OF COHERENT FSK

If we inspect Fig. 1, or (4), then it is evident that the phase at times that are odd (even) multiples of T will be odd (even) multiples of πh . Since all phases are modulo 2π , the case $h = \frac{1}{2}$ stands out, because then the phase can take only the two values $\pm \pi/2$ at odd times and only the two values 0 and π at even times.

Thus one may design a receiver to make a decision between the two permitted phase values. It will be shown that this is an optimal decision scheme, provided that the decision is made after observing not the signal in one time interval, but the real (imaginary) part of its pre-envelope in two time intervals: the one before and the one after the phase that we wish to estimate. This can be seen by elementary methods.

Let $\phi(0) = 0$ or π be the phase at $t = 0$ and let $\omega = \pi(f_1 + f_2)$. Then

$$u(t) = \begin{cases} + \exp j\left(\omega t \pm \frac{\pi t}{2T}\right), \\ - \exp j\left(\omega t \pm \frac{\pi t}{2T}\right), \end{cases} \quad \phi(0) = \begin{cases} 0, \\ \pi, \end{cases} \quad 0 \leq t \leq T. \quad (8)$$

The sign within the exponent is positive for space, negative for mark.

A similar result holds for $-T \leq t \leq 0$, except that the sign within the exponent is not necessarily the same in both

intervals; but this is immaterial, because we can form

$$\operatorname{Re} \left\{ \exp j\left(\pm \frac{\pi t}{2T}\right) \right\} = \cos \frac{\pi t}{2T}$$

regardless of whether mark or space is being sent in either interval. Consequently, in the given interval of duration $2T$

$$\operatorname{Re} \{u(t) \exp -j\omega t\} = \begin{cases} + \cos \frac{\pi t}{2T}, \\ - \cos \frac{\pi t}{2T}, \end{cases} \quad \phi(0) = \begin{cases} 0, \\ \pi, \end{cases} \quad -T \leq t \leq T. \quad (9)$$

The function $\operatorname{Re} \{u(t) \exp -j\omega t\}$ depends only on $\phi(0)$, not on any other $\phi(nT)$, thus all unwanted information has been removed. The decision between $\phi(0) = 0$ and $\phi(0) = \pi$ can be optimally made, based on the waveform received in $(-T, T)$, since the signals $\pm \cos(\pi t/2T)$, $(-T \leq t \leq T)$ are antipodal and contain each the energy of one bit [18]. The probability of error is therefore the same as that of a coherent binary receiver (or same energy per bit), i.e., as function of the SNR γ

$$P_{\text{FSK}} = \operatorname{erfc} \sqrt{2\gamma} \\ \doteq \frac{1}{\sqrt{2\pi} \gamma} \exp -\gamma.$$

This is over 3 dB better than the error probability of conventional FSK [15],

$$P_{\text{FSK}} = \frac{1}{2} \exp -\gamma/2.$$

The optimal receiver structure is a clocked matched filter, for instance implemented by an integrate-dump circuit in baseband, which performs

$$J(0) = \int_{-T}^T \cos \frac{\pi t}{2T} \operatorname{Re} \{u(t) \exp -j\omega t\} dt \quad (10)$$

and decides

$$\phi(0) = 0, \quad J(0) > 0$$

$$\phi(0) = \pi, \quad J(0) < 0.$$

Corresponding decisions in every second interval pair give $\phi(2nT)$ and this yields half the number of transmitted bits.

The other half comes from the imaginary part of u , where a similar relation holds for $\phi(T)$ as decided during the interval $(0, 2T)$,

$$J(T) = \int_0^{2T} \sin \frac{\pi t}{2T} \operatorname{Im} \{u(t) \exp -j\omega t\} dt \\ \phi(T) = \frac{\pi}{2}, \quad J(T) > 0 \\ \phi(T) = -\frac{\pi}{2}, \quad J(T) < 0 \quad (11)$$

and so on. Finally, the transmitted bit stream can be recovered from the resulting $\phi(nT)$.

No further effort to optimize the receiver is required; we can concentrate on implementing the mathematical receiver structure with practical circuits.

VI. SHIFTING INTO BASEBAND

$\operatorname{Re} \{u(t) \exp -j\omega t\}$ and $\operatorname{Im} \{u(t) \exp -j\omega t\}$ are obtained from the received signal

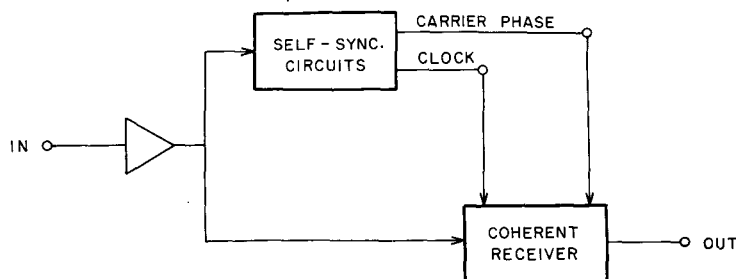


Fig. 3. Self-synchronizing receiver.

$$s(t) = \text{Re} \{u(t)\}$$

by beating it into baseband. This requires generating a phase-locked carrier at frequency $\omega/2\pi$, multiplying $s(t)$ with $\cos \omega t$ and $\sin \omega t$, respectively, in two balanced modulators, and then removing the second-harmonic terms from the two outputs, which by then are usually called I channel and Q channel.

The other operations of (10) and (11) are straightforward. They require at the receiver two auxiliary signals: the clock that is needed for integrating over an interval of length $2T$ and the precisely phase-locked carrier signals $\cos \omega t$ and $\sin \omega t$.

There are various ways to transport both signals, but the most elegant method is to regenerate them at the receiver by a self-synchronizing scheme, rather than wasting power and bandwidth for transmitting phase and clock reference signals.

Fig. 3 illustrates the overall schematic of a self-synchronizing receiver; after a preamplifier, the signals are split into two paths, one extracts the timing and phasing information, necessarily at a much slower rate than the signaling information [19]; and the other path contains a coherent detector of the signals, using the reference signals from the self-synchronizing subsystem.

VII. SELF-SYNCHRONIZATION OF THE RECEIVER

The self-synchronization of the fast FSK is most easily explained by reference to Sunde's FSK [6], [7], which has $h = 1.0$. Since h is a (normalized) frequency difference, a frequency-doubling circuit doubles h and thus transforms the fast FSK into Sunde's FSK.

Now Sunde's FSK has the stated disadvantage, that part of its power is in the spectral lines of the two carriers and thus not available for signaling. But we do not need the signaling—this is done at $h = \frac{1}{2}$; on the contrary we turn the spectral lines of Sunde's FSK into an advantage, since they are at $2f_1$ and $2f_2$ and therefore ideally suited to generate for the fast FSK the phase reference.

$$f_c = (2f_1 + 2f_2)/4$$

and the clock

$$1/T = 2f_2 - 2f_1.$$

In principle, this could be done with the circuit in Fig. 4.

A frequency doubler feeds two phase-locked loops (PLL), which extract $2f_1$ and $2f_2$, and send them to a balanced modulator.

The low-frequency output $2f_2 - 2f_1$ of the modulator is the clock, the high-frequency output $2f_1 + 2f_2$ is divided by 4 to generate the two phase reference signals. All the reference signals can be generated practically error free, by narrowing the PLL bandwidth, provided that sufficient time is available for initial synchronization.

The phase and clock signals are then used for a conventional coherent demodulation and detection of the signals in the

main path, which do not pass through the doubler stage, but remain at $h = \frac{1}{2}$; and this, in principle, is the receiver.

If certain periodic signals are used as modulation, then the frequency doubler may generate a line at $2f_c = f_1 + f_2$. This frequency should not be used in the self-synchronizing circuit, since a spectral line at $2f_c$ is in general not available. Some care is necessary to prevent false locking of the PLL if they should find $2f_c$ in the presence of such a periodic modulation.

A different problem is the possible loss of lock if a long string of only mark (or only space) is received. Some improvement can be obtained by modulating the remaining line with the clock to regenerate the missing line, but this works only as long as the clock itself does not drift out of sync. This problem is common to all FSK. The best remedy, if available, is suitable source encoding to prevent such long strings from occurring.

VIII. 90° PHASE AMBIGUITY

The divide-by-four circuit generates a phase ambiguity of the reference phases by some multiple of 90°. Unlike the 4-phase PSK, the fast FSK can resolve this ambiguity, for instance by comparing the coherent output with a bit stream from the frequency doubler. This bit stream is a conventional FSK, except with some noise added (by the doubling circuit) and it is not desirable to use it directly in the receiver, but it still contains enough good bits to resolve the ambiguity if comparison is made over several time intervals.

An alternative method is easier to implement. We treat the ambiguity in two ways. The 90° ambiguity can be removed by a more careful layout of the self-synchronizing circuit, keeping in mind that we really have a binary system and should not allow fourth-order effects to cause phase ambiguities. Fig. 5 illustrates the circuitry.

The circuit of Fig. 5 has the same doubler, PLL, and master clock as the circuit of Fig. 4, but it generates the phase references differently. The outputs of the PLL are divided by two and then added or subtracted. Since the phase at the divider output is ambiguous by 180°, we do not know which adder gives the sum and which the difference, but this does not matter since the outputs are, respectively, $\pm \cos \omega t \cos \pi t/2T$ and $\pm \sin \omega t \sin \pi t/2T$ and both are the required reference signals, when they are gated between the nulls of their respective modulation.

IX. 180° PHASE AMBIGUITY

The only ambiguity left is the $\pm$ sign in front of the generated reference and this can be taken care of by the method of differential encoding, as is usually done in coherent binary PSK.

Therefore we relate the transmitted bit stream $a_n = \pm 1$, $n = 1, 2, \dots$ to the received signals in the I and Q channels. First we notice that because of the 180° ambiguities only the phase differences $\phi[(n-1)T] - \phi[nT]$ can be used. Inspection of Fig. 1 shows that a sequence mark-space or space-mark leaves ϕ unchanged, while a sequence mark-mark

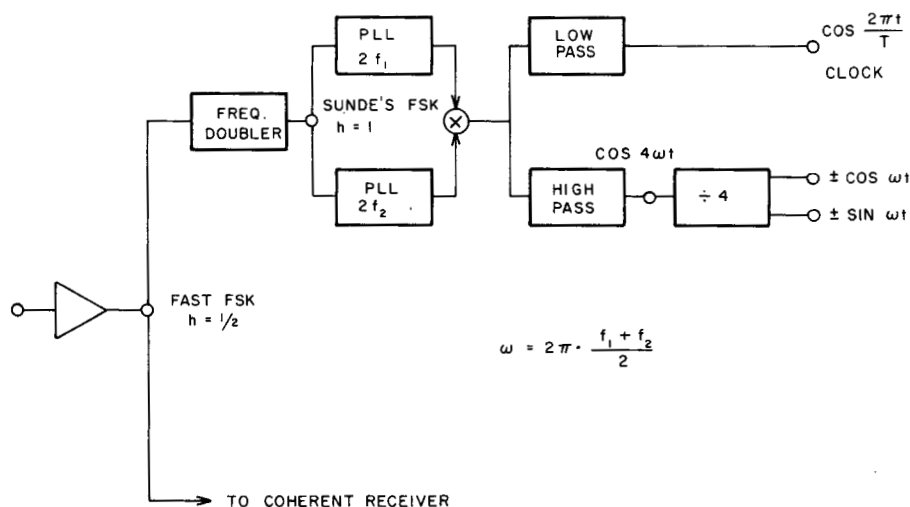


Fig. 4. Generation of clock and phase reference.

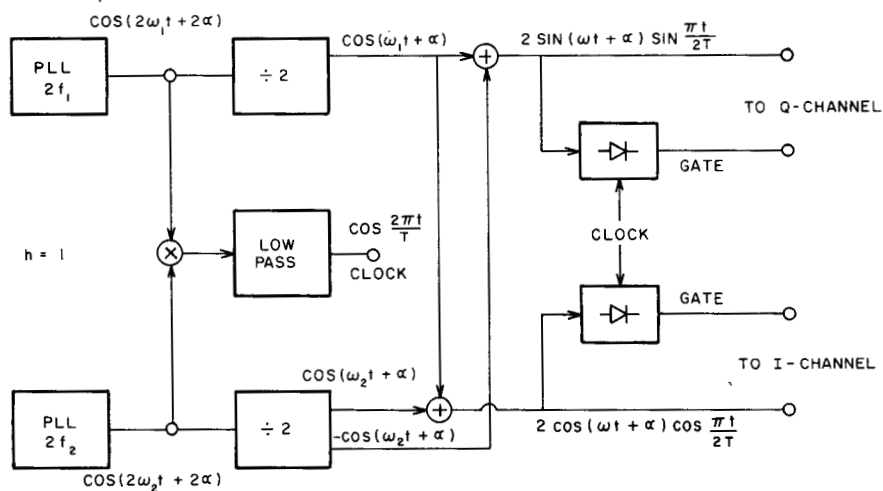


Fig. 5. Alternative generation of references from the PLL outputs of Fig. 4

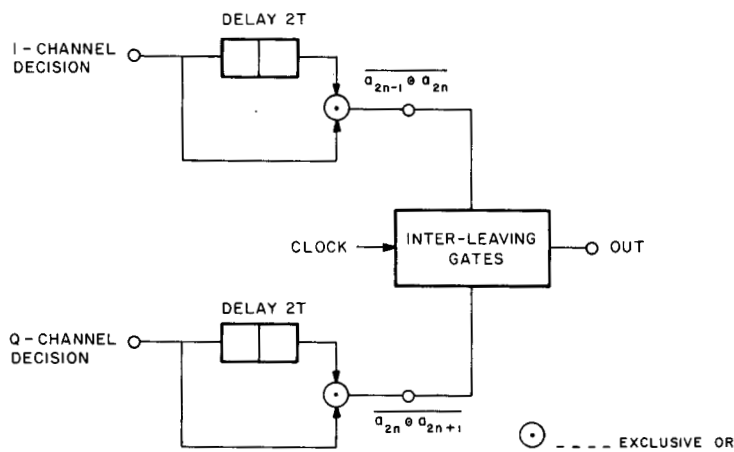


Fig. 6. Digital outputs when $\{a_n\}$ was transmitted.

or space-space changes ϕ by 180° after 2 bit intervals. Thus, at time $t = 2nT$, a transition at the input corresponds to no transition at the output of the I channels and at time $t = (2n + 1)T$, a transition at the input corresponds to no transition in the Q channel, and vice versa.

The circuit of Fig. 6 shows the various logic operations that recover the differentially encoded input from the outputs of the I and Q channels.

The circuit decodes the differentially encoded input $a_n \cdot a_{n+1}$ when a_n was sent.

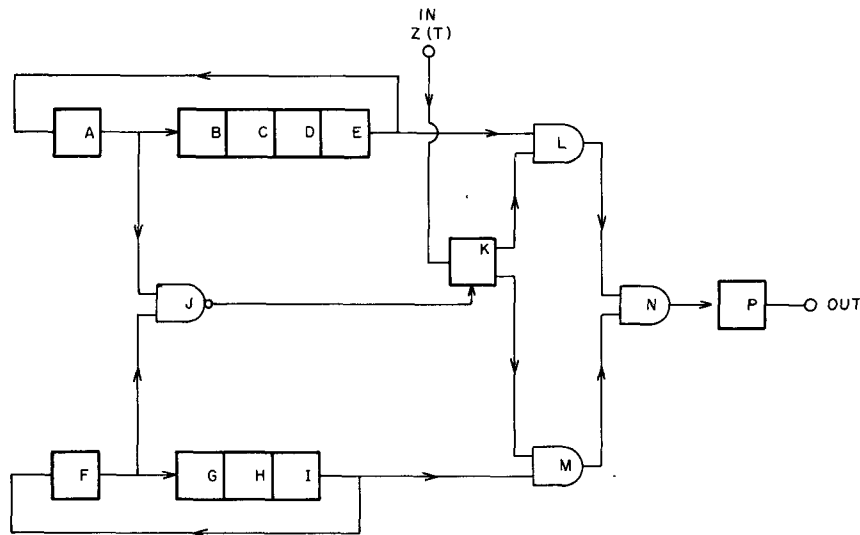


Fig. 7. Generator for fast FSK.

X. EFFECT OF FILTERING AND INTERSYMBOL INTERFERENCE

If the fast FSK signals are passed through a narrow-band filter, at the transmitter and/or at the receiver front end, then the signals available for the phase decisions will no longer be independent. Two types of interference may occur: 1) within each channel and 2) between the *I* and the *Q* channel.

The second type can be avoided by careful circuit designs. There will be no such interference if the overall filter action is that of a symmetrical bandpass,

$$G(f_c + f) = G^*(f_c - f)$$

because such a bandpass has a weighting function

$$g(t) \cdot \exp - 2\pi j f_c t,$$

where $g(t)$ is a real function.

The bandpass action is then the correlation of $\exp j\phi(t)$ with $g(t)$ and this leaves the real part of $\exp j\phi(t)$ real and the imaginary part imaginary, so that no interference between the *I* channel and the *Q* channel occurs.

The intersymbol interference within the *I* (or *Q*) channel cannot be removed so easily. It limits the speed with which we can transmit bits through a given system. But, in each of the two baseband channels, bits occur at half-speed ($1/2T$) and the signaling function is continuous, since switching occurs only at the nulls of $\cos \pi t/2T$. Thus the intersymbol interference will be considerably smaller than when rectangular pulses are transmitted in an IF bandwidth of $1.2/T$.

When the IF bandwidth is $0.75/T$, then the intersymbol interference of the fast FSK is still negligible. Higher bit rates are possible by making the intersymbol correction by a transversal filter [20], [21] at baseband, which is by now a well-established technique. Using only an intersymbol correction before decomposition into *I* and *Q* channel seems to give poorer results [22].

The output of the narrow-band filter will show amplitude fluctuations, mainly increases near the mark-space transition. This is unavoidable. If one tries to remove the amplitude by a limiter, then the sidebands are regenerated and the effective filter bandwidth is widened [12]. There is also the usual small reduction of SNR due to hard limiter action, but the probability of error does not increase significantly. The decision is based upon the quadrant in which the received signal is in a given interval. If we now make no intersymbol interference correction, but ideally investigate per interval only one sample of signal plus noise plus interference, then the decision

made from this sample, and hence the error probability, is not affected by hard limiting after the filter.

XI. DIGITAL GENERATION OF THE FAST FSK SIGNAL

When generating the fast FSK, for good performance it is required not only to generate a coherent FSK, but also to hold the normalized frequency excursion h to precisely the value of $h = \frac{1}{2}$.

This can be achieved by a suitable feedback circuit around a conventional coherent FSK; or by a circuit that generates two sine waves at f_1 and $f_2 = f_1 + h/T$, each with positive and negative polarity, and a logic circuit, which, on frequency shifting, selects that polarity of the new frequency that maintains phase coherence.

A more elegant generator of the fast FSK can be constructed as follows.

First Sunde's FSK [6] is generated at IF, then sent through a flip-flop, which divides by 2; then the harmonics of the resulting square wave are removed by a zonal filter and the resulting signal is the fast FSK, since Sunde's FSK has a normalized frequency shift of $h = 1$, which, after the frequency divider stage, becomes $h = \frac{1}{2}$.

If the desired frequency range allows it, then Sunde's FSK can be obtained in an all-digital circuit as follows.

Let n be some small integer. Generate a master clock with frequency $n(1 + 1)/T$.

Count down to derive three pulse trains, the first with n/T ; the other with $(n + 1)/T$ pulses, and the third train with $1/T$ pulses. This third train serves as clock for the bit-stream encoder.

In each time interval T we select n pulses from the first train if we transmit mark or $(n + 1)$ pulses from the second train, if we transmit space. This can be done by a digital circuit under control of the master clock. The fundamental of the resulting pulse train is then an FSK with $h = 1$; but it is not necessary to obtain the fundamental because the train itself is ideally suited to drive a divide-by-two flip-flop. The fundamental at the flip-flop output is then the fast FSK.

Fig. 7 illustrates a digital circuit that implements this scheme for $n = 4$.

XII. CONCLUSION

The fast FSK is a binary FSK with these priorities.

- 1) It is phase coherent.
- 2) It has a low deviation ratio, $h = \frac{1}{2}$.
- 3) It occupies a small bandwidth.

4) It uses a phase detector as demodulator. First, with the help of a modification of the Costas receiver [23], self-synchronization of phase and clock is achieved; then the signals are coherently demodulated and detected.

Such a receiver shows a distinct improvement over the bandwidth and SNR required for conventional reception of binary FSK signals. For instance, when no correction for intersymbol interference is used, then the fast FSK can be transmitted in an RF bandwidth of only 0.75 times the bit rate, yet received with about 3 dB less power than conventional FSK at equal bit rate and error probability.

The theory of the scheme is fairly simple, once it is recognized where the information, which makes this improvement possible, is to be found; and the implementation is straightforward.

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Equivalence of Time-Multiplexed and Frequency-Multiplexed Signals in Digital Communications

UZI TIMOR

Abstract—In comparing different techniques for multiplexing N binary data signals into a single channel, time-division multiplexing

(TDM) is known to have a theoretic efficiency of 100 percent (neglecting sync power) and thus seems to outperform frequency-division multiplexing systems (FDM). By considering more general FDM systems, we will show that both TDM and FDM are equivalent and have an efficiency of 100 percent. The difference between the systems is in the multiplexing and demultiplexing subsystems, but not in the performance or in the generated waveforms.

INTRODUCTION

Let $d_k(t)$, $k = 1, \dots$, and N be N binary data signals to be transmitted coherently over a single RF channel by phase modulating the carrier. The transmitted signal is

$$s(t) = \sqrt{2P} \sin(\omega_c t + \theta(t)), \quad (1)$$

where P is the total power, ω_c is the carrier frequency, and $\theta(t)$ is generated by the data according to the type of modulation used.

In order to compare the efficiency of various methods of modulation, we will assume that synchronization is achieved without loss of power, i.e., without allocating a fraction of the total power to a special sync channel, unmodulated RF or subcarrier, or frame sync. In doing so, we can ignore secondary effects due to the particular scheme of sync or detection and base the comparison on the theoretic efficiencies of the various modulation schemes.

Previous work [1], [2] of the author considered phase-shift-keying/phase modulation (PSK/PM) systems, where the data signals are frequency multiplexed with subcarriers and combined to phase modulate the carrier. It was shown that: square-wave subcarriers are superior to sine-wave subcarriers and that using biphas-modulated square-wave subcarriers $sq(\omega_k t)$, the optimum PSK/PM configuration is one of the following.

1) The conventional scheme

$$\theta(t) = \sum_{k=1}^N \theta_k a_k(t). \quad (2)$$

2) The interplex scheme [3]

$$\theta(t) = \theta_1 a_1(t) + \sum_{k=2}^N \theta_k a_k(t) a_1(t), \quad (3)$$

where

$$a_k(t) = d_k(t) sq(\omega_k t) \quad (4)$$

are data-modulated subcarriers and the θ_k are the modulation angles that determine the power allocation to the data channels. The desirability of one of these schemes over the other depends on the design parameters, that is, the unmodulated RF power required for tracking the phase and the ratio of powers in the data channels.

In general, the interplex method is superior to the conventional scheme for small N . However, for $N \geq 3$ both schemes operate at an efficiency that is strictly smaller than 100 percent and decreases monotonically with N .

By considering more general FDM systems, we will show that an efficiency of 100 percent can be achieved for all N . Specifically, by using four-phase-modulated (instead of biphas-modulated) square-wave subcarriers (FPSK) in the interplex mode we can eliminate any power loss due to cross-modulation products. Moreover, TDM of two data channels and FPSK/interplex not only have the same performance, but they are actually equivalent. That is, any waveform that is the outcome of time-division multiplexing can be written as an FPSK signal and vice versa. The commutation rates in TDM are related to the subcarrier frequency in FDM systems and the

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The author was with the Jet Propulsion Laboratory, Pasadena, Calif. He is now with the Scientific Department, Israel Ministry of Defence, Hakiry, Israel.